

Year 11

Mathematics

Assessment Task 3

2015

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- In Questions 11 - 14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II

60 marks

- Attempt Questions 11-14
- Start each question in a new writing booklet
- Write your name on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your name and "N/A" on the front cover
- Allow about 1 hour 45 minutes for this section

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

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Section I

10 marks

Attempt Questions 1–10

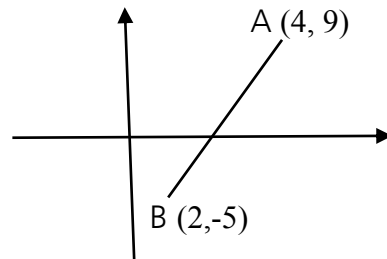
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 A is the point (4, 9) and B is the point (2, -5).

Which of the following gives the gradient of AB:

- (A) 7
- (B) -7
- (C) 2
- (D) 5



- 2 Find θ if $\sin\theta = \frac{\sqrt{3}}{2}$ where $0 \leq \theta \leq 360^\circ$

- (A) $\theta = 60^\circ$ and 120°
- (B) $\theta = 60^\circ$ and 240°
- (C) $\theta = 240^\circ$ and 300°
- (D) $\theta = 120^\circ$ and 240°

3 What is the exact value of $\cos 135^\circ + \operatorname{cosec} 60^\circ$?

(A) $\frac{2\sqrt{2}-\sqrt{3}}{\sqrt{6}}$

(B) $\frac{2\sqrt{2}-1}{\sqrt{2}}$

(C) $\frac{2\sqrt{2}+\sqrt{3}}{\sqrt{6}}$

(D) $\frac{2\sqrt{2}+1}{\sqrt{2}}$

4 Calculate the size of each interior angle of a regular polygon with 20 sides.

(A) 200°

(B) 180°

(C) 162°

(D) 18°

5 Solve $(2^x)^2 = 8$

(A) $x = 2$

(B) $x = 4$

(C) $x = 2\sqrt{2}$

(D) $x = \frac{3}{2}$

6 Find $\frac{dy}{dx}$ if $y = \frac{2}{x^4}$

(A) $\frac{-8}{x^5}$

(B) $\frac{-8}{x^3}$

(C) $\frac{-10}{x^8}$

(D) $\frac{-6}{x}$

7 Find $f'(2)$ if $f(x) = (3x - 4)^3$

(A) 72

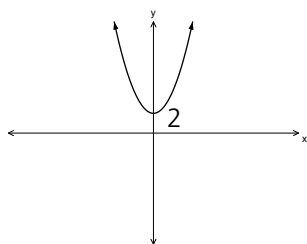
(B) 36

(C) 8

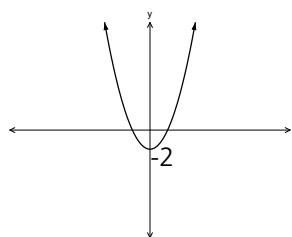
(D) 27

8 Which of the following graphs best represents $y = (x + 2)^2$

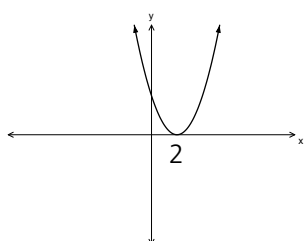
(A)



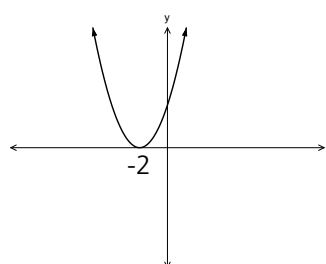
(C)



(B)



(D)



- 9 Let $f(x) = \sqrt{4-x^2}$. What is the range of $f(x)$?
- (A) $-2 \leq y \leq 2$
- (B) $0 \leq y \leq 4$
- (C) $0 \leq y \leq 2$
- (D) $0 \leq x \leq 2$
- 10 The tangent to $y = x^2 + x$ at $x = a$ is parallel to the line $y = 5 - 3x$.
The value of a is:
- (A) -2
- (B) -3
- (C) 1
- (D) $\frac{3}{2}$

End of Section 1

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

Marks

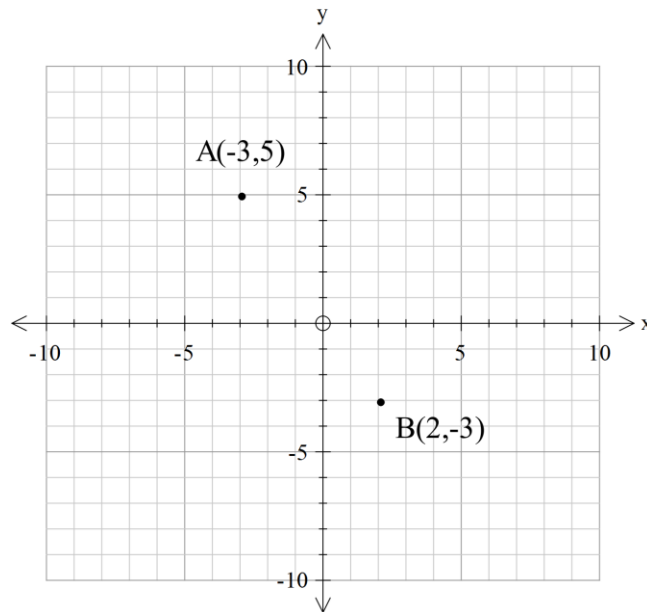
- (a) By rationalising the denominator, express $\frac{8}{3-\sqrt{5}}$ in the form $a + b\sqrt{5}$. 2
- (b) The fuel tank on my new car was 40% full. Jiya added 28 litres and then found it was 75% full. How much fuel does the tank hold? 1
- (c) Solve $\frac{3x-2}{5} = \frac{x}{4}$ 2
- (d) Simplify $\frac{a+b}{4x-2y} \times \frac{2x-y}{a^2-b^2}$ 3
- (e) Solve the inequality $|2x+3| < 9$ and show the solutions on a number line. 3
- (f) Solve the equations $y = 5x - 1$ and $y = x^2 + 3x$ simultaneously. 3
- (g) Evaluate $|30 - 12 \times 3| - |-30 + 12 \times 3|$ 1

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

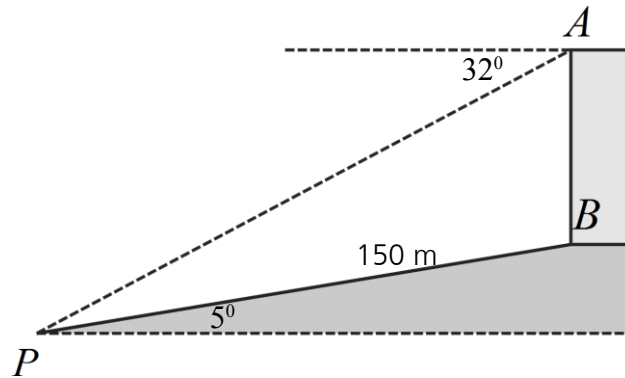
Marks

- (a) An interval AB is formed by the points A(-3, 5) and B(2, -3) on the number plane shown below:



- | | | |
|-------|---|---|
| (i) | Find the exact length of the interval AB | 1 |
| (ii) | Find the gradient of the interval AB. | 1 |
| (iii) | Show that the equation of the line AB is $8x + 5y - 1 = 0$. | 2 |
| (iv) | Find the equation of the line through C(5, 8) perpendicular to AB. | 2 |
| (v) | Calculate the perpendicular distance of the point C from the line AB. | 2 |
| (vi) | Calculate the area of the triangle ABC. | 1 |

- (b) A point P is at the foot of the hill which is inclined at 5° to the horizontal. The base B, of a tower AB, is situated 150 metres up the incline of the hill from P. From the top of the tower the angle of depression of the point P is 32° .



- (i) Find the size of $\angle APB$ and of $\angle PAB$. 2
- (ii) Find the height of the tower, AB, correct to the nearest metre. 2
- (c) Simplify $\frac{1}{\operatorname{cosec}^2 \theta} + \frac{1}{\sec^2 \theta} + \frac{1}{\cot^2 \theta}$ 2

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) Sketch the following graphs and clearly label the vertex, asymptotes (if any) and the points where the graph cuts the x and y -axes.

(i) $y = x^2 + 2x - 3$ 2

(ii) $y = \frac{3}{x+2}$ 2

- (b) Use the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find the derivative of $f(x) = x^2 - 5x$. 2

- (c) Differentiate the following functions:

(i) $f(x) = 5 - 7x + 13x^4$ 1

(ii) $f(x) = \frac{2x-3}{3x-2}$ 2

(iii) $f(x) = (3x^2 - 1)(x - 1)^3$ 3

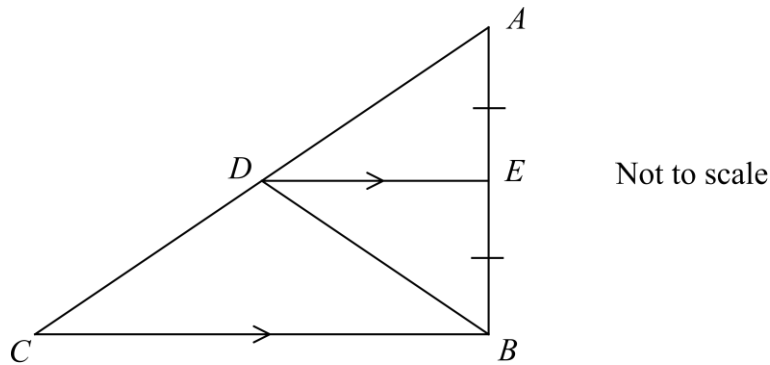
- (d) Find the equation of the tangent to the curve $y = 2x^2 - 4x + 5$ at the point where $x = 3$. 3

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

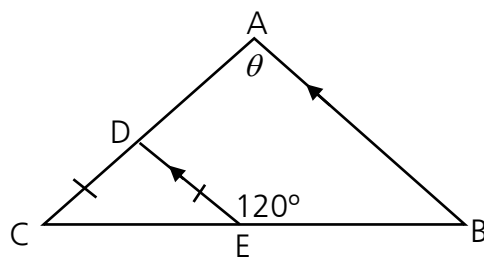
Marks

- (a) In the diagram below $\triangle ABC$ has a right angle at B. Point E is the midpoint of AB. Point D lies on AC and DE is parallel to BC.



- | | |
|--|---|
| (i) Explain why $\angle AED$ is a right angle. | 1 |
| (ii) Prove that $\triangle AED \equiv \triangle BED$ | 2 |
| (iii) Show that $BD = DC$ | 3 |

- (b) In the diagram shown $DE \parallel AB$ and $CD = DE$



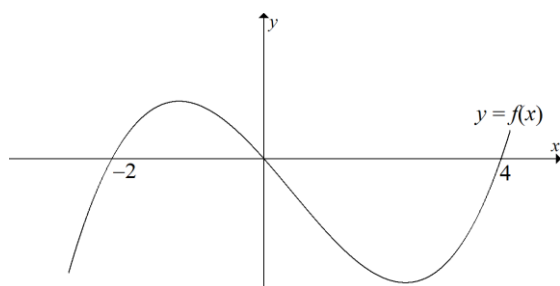
Find $\angle \theta$ giving reasons. 3

- | | |
|--|---|
| (c) Find the centre and radius of the circle $x^2 + y^2 - 2x + 4y + 1 = 0$. | 2 |
|--|---|

- | | |
|--|---|
| (d) Determine if the function $f(x) = \frac{x}{x^2 - 1}$ is odd or even. | 2 |
|--|---|

- (e) Copy the graph of $y = f(x)$ shown below in your answer booklet and draw a sketch of the gradient function, $y = f'(x)$ on the same diagram.

2



End of Examination

Student Name:

2015 Year 11 Assessment Task 3

Mathematics

Section 1 Multiple-Choice Answer Sheet

1 A ☐ B ☐ C ☐ D ☐

2 A ☐ B ☐ C ☐ D ☐

3 A ☐ B ☐ C ☐ D ☐

4 A ☐ B ☐ C ☐ D ☐

5 A ☐ B ☐ C ☐ D ☐

6 A ☐ B ☐ C ☐ D ☐

7 A ☐ B ☐ C ☐ D ☐

8 A ☐ B ☐ C ☐ D ☐

9 A ☐ B ☐ C ☐ D ☐

10 A ☐ B ☐ C ☐ D ☐

YEAR 11 AT3 2015 SOLUTIONS

SECTION 1

1. A 2. A 3. A 4. C 5. D 6. A
7. B 8. D 9. C 10. A

SECTION 2

$$a) \quad \frac{8}{3-\sqrt{5}} = \frac{8}{3-\sqrt{5}} \times \frac{3+\sqrt{5}}{3+\sqrt{5}} \quad \checkmark$$

$$= \frac{24+8\sqrt{5}}{9-5}$$

$$= \frac{8(3+\sqrt{5})}{4}$$

$$= \underline{2(3+\sqrt{5})} \text{ or } 6+2\sqrt{5} \quad \checkmark$$

b) let tank hold x L

$$\therefore 0.4x + 28 = 0.75x$$

$$0.35x = 28$$

$$\underline{x = 80 \text{ L}} \quad \checkmark$$

$$c) \frac{3x-2}{5} = \frac{x}{4}$$

$$\cancel{20}^4 \left(\frac{3x-2}{5} \right) = \left(\frac{x}{\cancel{4}} \right) \cancel{20}^5$$

$$12x - 8 = 5x$$

$$7x = 8$$

$$x = \frac{8}{7}$$

correct progress
to step 3.

$$d) \frac{a+b}{4x-2y} \times \frac{2x-y}{a^2-b^2}$$

$$= \frac{\cancel{(a+b)}}{2(\cancel{2x-y})} \times \frac{\cancel{(2x-y)}}{(a+b)(a-b)}$$

$$= \frac{1}{2(a+b)}$$

✓ correct factorisation

✓ correct factors
✓ correct simplifying

$$e) |2x+3| < 9$$

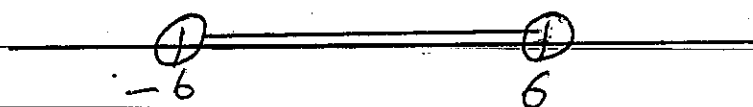
$$2x+3 < 9 \quad \text{OR} \quad -(2x+3) < 9$$

$$\underline{2x < 6} \quad \text{OR} \quad 2x+3 > -9$$

✓

$$2x > -12$$

$$\underline{x > -6} \quad \checkmark$$



✓

$$f) \quad y = 5x - 1$$

$$y = x^2 + 3x$$

$$\therefore x^2 + 3x = 5x - 1$$

$$x^2 - 2x + 1 = 0$$

$$(x-1)^2 = 0$$

$$x-1 = 0$$

$$x = 1 \Rightarrow y = 5(1) - 1 = 4$$

$$\therefore (1, 4) \quad \checkmark \text{ both } x \text{ \& } y \text{ values.}$$

$$g) \quad |30 - 12 \times 3| - |-30 + 12 \times 3|$$

$$= |30 - 36| - |-30 + 36|$$

$$= |-6| - |6|$$

$$= 6 - 6$$

$$= 0$$

✓

Question 12

$$a) \quad (i) \quad AB = \sqrt{(-3-2)^2 + (5--3)^2} \\ = \sqrt{89} \quad \checkmark$$

$$(ii) \quad m_{AB} = \frac{-3-5}{2--3} \\ = -\frac{8}{5} \quad \checkmark$$

$$(iii) \quad m_{AB} = -\frac{8}{5} \quad B(2, -3)$$

$$y--3 = -\frac{8}{5} (x-2) \quad \checkmark$$

$$5(y+3) = -8(x-2)$$

$$5y + 15 = -8x + 16$$

$$8x + 5y - 1 = 0 \quad \text{as required.} \quad \checkmark$$

$$(iv) \quad m_{AB} = -\frac{8}{5} \Rightarrow m_{\perp} = \frac{5}{8} \quad C(5, 8)$$

$$\therefore y-8 = \frac{5}{8} (x-5) \quad \checkmark$$

$$8y - 64 = 5x - 25$$

$$\underline{\underline{5x - 8y + 39 = 0}} \quad \checkmark$$

$$(v) AB: 8x + 5y - 1 = 0 \quad C(5, 8)$$

$$d_1 = \frac{|ax + by + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|(8)(5) + (5)(8) - (-1)|}{\sqrt{8^2 + 5^2}} \quad \checkmark$$

$$= \frac{79}{\sqrt{89}} \quad \checkmark$$

$$(vi) A = \frac{1}{2} \times \frac{79}{\sqrt{89}} \times \sqrt{89}$$

$$= \frac{79}{2} \text{ m}^2 \quad \checkmark$$

b) (i)

$$\angle APB = 32 - 5 = \underline{\underline{27^\circ}}$$

$$\angle PAB = 90 - 32 = \underline{\underline{58^\circ}} \quad \checkmark \text{ Both correct.}$$

$$(ii) \frac{h}{\sin 27} = \frac{150}{\sin 58}$$

$\checkmark$ COA acceptable.

$$h = \frac{150 \sin 27}{\sin 58}$$

$$= \underline{\underline{80 \text{ m}}} \text{ (correct to nearest m)} \quad \checkmark$$

$$c) \frac{1}{\operatorname{cosec}^2 \theta} + \frac{1}{\sec^2 \theta} + \frac{1}{\cot^2 \theta}$$

$$= \sin^2 \theta + \cos^2 \theta + \tan^2 \theta$$

$$= 1 + \tan^2 \theta$$

$$= \underline{\sec^2 \theta}$$

✓
(correct either
step 2 or 3.)

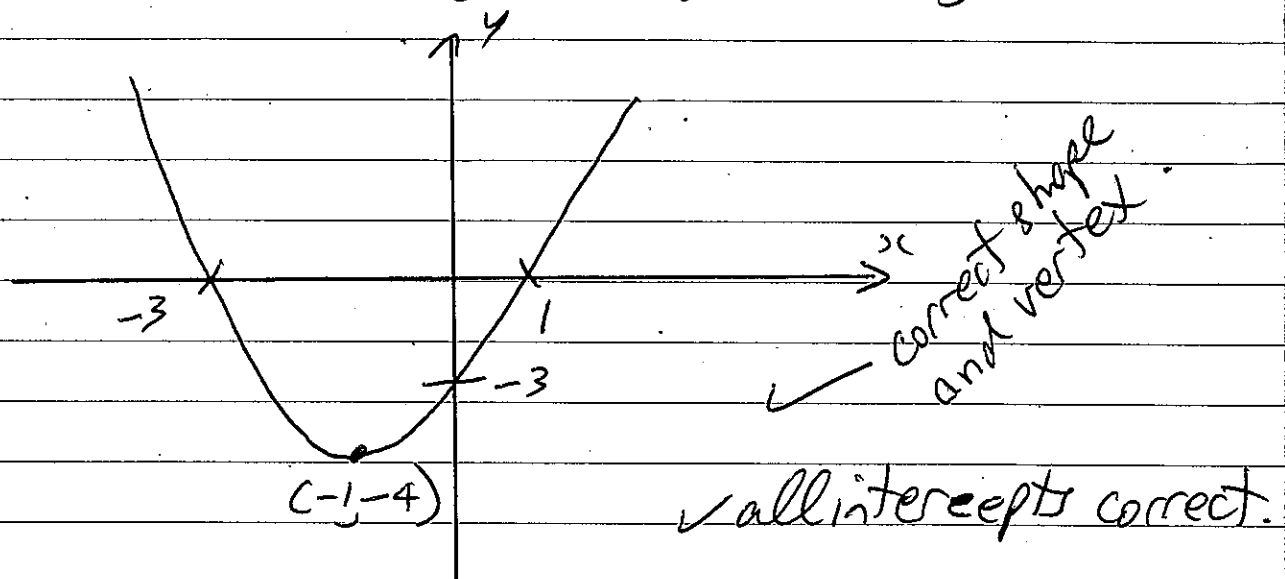
✓

Question 13

a) (i) $y = x^2 + 2x - 3$

$$= (x+1)^2 - 4$$

$$\therefore V(-1, -4) \quad (0, -3) \quad (1, 0) \quad (-3, 0)$$

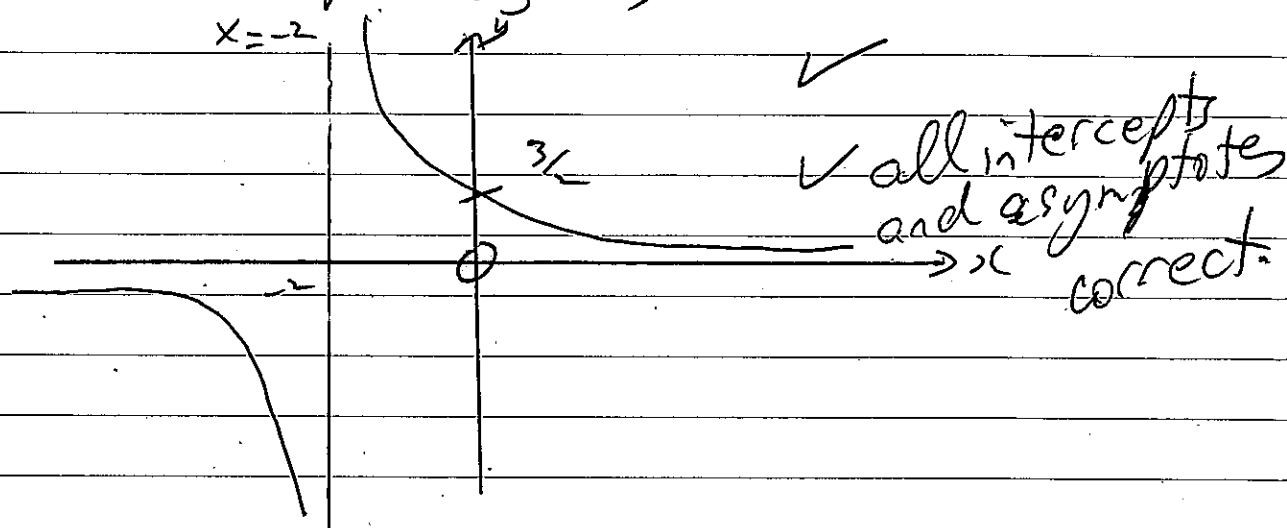


(ii) $y = \frac{3}{x+2} \Rightarrow y(x+2) = 3$

Vertical asymptote : $x = -2$

horizontal " : $y = 0$

y-intercept $(0, \frac{3}{2})$



$$b) f(x) = x^2 - 5x$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 5(x+h) - (x^2 - 5x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 5x - 5h - x^2 + 5x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h - 5)}{h}$$

$$= 2x - 5$$

no mark if
this method is
not used.

$$c) f(x) = 5 - 7x + 13x^4$$

$$(i) f'(x) = -7 + 52x^3$$

$$(ii) f(x) = \frac{2x-3}{3x-2}$$

$$f'(x) = \frac{2(3x-2) - (3)(2x-3)}{(3x-2)^2}$$

$$= \frac{6x - 4 - 6x + 9}{(3x-2)^2}$$

$$= \frac{5}{(3x-2)^2}$$

(quotient rule)
OK if product
rule is applied

$$c) (iii) f(x) = (3x^2 - 1)(x - 1)^3$$

$$\text{Let } u = 3x^2 - 1 \Rightarrow u' = 6x$$

$$\text{and } v = (x - 1)^3 \Rightarrow v' = 3(x - 1)^2 \quad \begin{array}{l} \checkmark \\ u' \checkmark \\ v' \checkmark \\ \text{correct} \end{array}$$

$$\therefore f'(x) = (3x^2 - 1)(3)(x - 1)^2 + (x - 1)^3(6x) \quad \checkmark$$

$$= 3(x - 1)^2 [3x^2 - 1 + 2x(x - 1)] \quad \checkmark$$

$$= 3(x - 1)^2 (5x^2 - 2x - 1)$$

d)

$$y = 2x^2 - 4x + 5$$

$$x = 3 \Rightarrow y = 2(3)^2 - 4(3) + 5 = 11$$

$$m_T = \frac{dy}{dx} = 4x - 4$$

$$x = 3 \Rightarrow m_T = 4(3) - 4 = 8$$

$\checkmark$
correct point
and gradient

$$m_T = 8 \quad (3, 11)$$

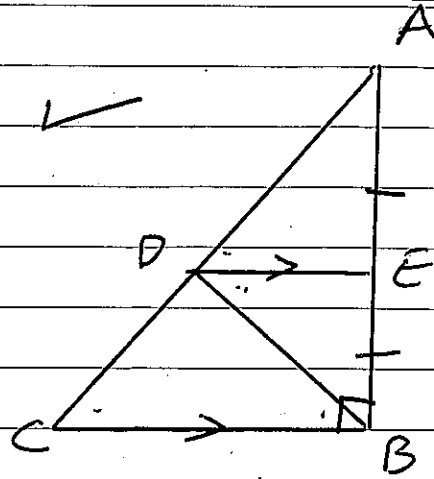
$$\therefore \text{equation of tangent: } y - 11 = 8(x - 3) \quad \checkmark$$

$$y - 11 = 8x - 24$$

$$\underline{\underline{8x - y - 13 = 0}} \quad \checkmark$$

Question 14

a) (i) $\angle AED = 90^\circ$ ✓
Corresponding $\angle$ s $DE \parallel CB$



(ii) In $\triangle AED$ and $\triangle BED$

$$\begin{aligned}\angle AED &= \angle BED \\ &= 90^\circ \text{ (AEB is a straight line)}\end{aligned}$$

ED is common ✓
AE = BE (given)

$$\therefore \triangle AED \equiv \triangle BED \text{ (SAS)} \quad \checkmark$$

(iii) To show $BD = DC$

$$BD = AD \text{ (Corresponding sides of } \equiv \Delta \text{s)}$$

$$\angle EDB = \angle DBC \text{ (Alternate } \angle \text{s } DE \parallel CB)$$

$$\angle ADE = \angle ACB \text{ (Corresponding } \angle \text{s } DE \parallel CB)$$

$$\text{But } \angle EDB = \angle EDA \text{ (Matching } \angle \text{s of } \equiv \Delta \text{s)}$$

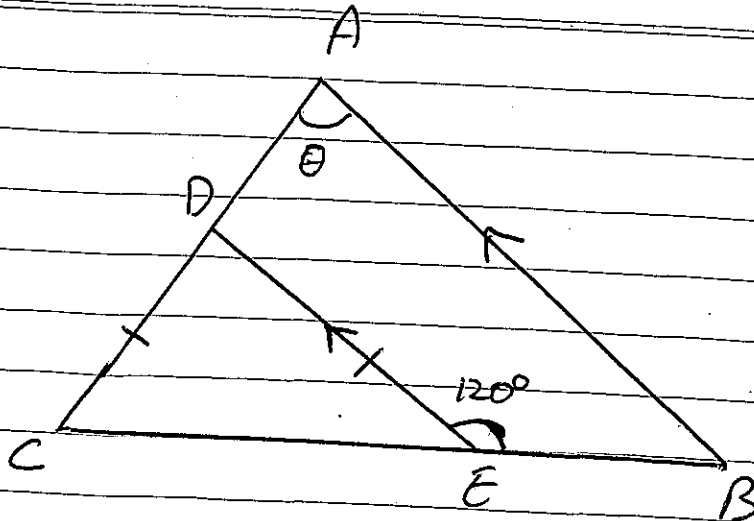
$$\therefore \angle DBC = \angle ACB$$

$\therefore \triangle DBC$ is isosceles (base $\angle$ s equal)

$$\therefore CD = BD = AD$$

$\therefore BD = AD$ as required.

b)



$$\angle DEC = 60^\circ \text{ (straight line)}$$

$$\angle ACB = 60^\circ \text{ (base } \angle \text{ of isosceles } \Delta)$$

$$\therefore \angle CDE = 180 - 120 \text{ (}\angle \text{ sum of } \Delta)$$

$$= 60^\circ$$

$$\therefore \theta = 60^\circ \text{ (corresponding } \angle \text{ AB} \parallel \text{DE)}$$

c) $x^2 - 2x + y^2 + 4y = -1$

$$(x-1)^2 + (y+2)^2 = -1 + 1 + 4$$

$$= 4$$

$$\therefore C(1, -2) \text{ Radius} = 2$$

✓
some working.

d) $f(x) = \frac{x}{x^2-1}$

$$f(-x) = \frac{-x}{(-x)^2-1} = -\frac{x}{x^2-1}$$

$$-f(x) = -\frac{x}{x^2-1}$$

$$\therefore f(-x) = -f(x) \text{ Hence odd function.}$$

Question 14

e)

